MAT150C: Modern Algebra Lecture Notes

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2026-04-05

Contents

1 Lecture 1 and 2 **2**
1.1 Keywords 2
1.2 Content 2

2 Lecture 3 **3**
2.1 Keywords 3
2.2 Content 4

1 Lecture 1 and 2

1.1 Keywords

Representations

1.2 Content

Motivation - In general, groups can be found in numerous settings, both as purely abstract mathematical objects and as applied realizations. In pure mathematical contexts, we discuss the concept of group action on *sets*. Now, we extend the general action onto more structured mathematical spaces.

Definition 1. Let G be a group. An n-dimensional ***representation*** of G is a homomorphism

$$\rho : G \to GL_n(\mathbb{C})$$

Remark 1. The use of the general linear group allows for the action of groups on vector spaces. Note that $\mathbb{C}$ need not be the specified field of GL_n but it is the easiest to use.

Example 1. Let $G = C_m = \langle x \rangle$, where C_m is the cyclic group of order m. The map

$$\rho : C_m \to GL_1(\mathbb{C}) = \mathbb{C}^\times$$

is a representation. Note that under ρ (and recalling that x^m is the identity of C_m)

$$\underbrace{\rho(x)^m = \rho(x^m)}_{\text{homo prop}} = \rho(e) = 1$$

While representations can sometimes be trivial or obvious, they are not necessarily always easy to find. Furthermore, we must prove that ρ is a homomorphism. The following example provides more insight onto these hardships.

Example 2. Consider the dihedral group D_6 such that

$$\rho : D_6 \to GL_n(\mathbb{C}), \quad \rho(x) = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \text{ and } \rho(y) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Recall also that (* $x \sim$ *rotate and* $y \sim$ *reflect**)

$$D_6 = \langle x, y \mid y^2 = x^3 = (xy)^2 = e \rangle$$

To show that this is a valid representation, we verify that the homomorphism properties are in agreement with the relations imposed by the group elements i.e we must show that $\rho(x)^3, \rho(y)^2, (\rho(x)\rho(y))^2 = e \in GL_n(\mathbb{C})$. In doing so, we guarantee the representation behaves accordingly.

$$\begin{aligned} \rho(x)^3 &= \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}^3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \rho(y)^2 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ (\rho(x)\rho(y))^2 &= \left(\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right)^2 = \left(\begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} \right)^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Thus, $\rho(x), \rho(y)$ provide a representation of D_6.

Question - Suppose G was a group, not finitely generated, how would we go about finding a representation?

2 Lecture 3

2.1 Keywords

Representation Isomorphism, Irreducible, Group Actions, Vector Space Basis

2.2 Content

Recall the following definition

Definition 2. Let G be a group and S a finite set. The ***left group action*** of G on S is defined as the function

$$G \times S \to S, \quad (g, s) \mapsto g \cdot s$$

that satisfies

1. $g \cdot (h \cdot s) = (gh) \cdot s$
2. $e \cdot s = s$

Generally, common vector spaces are given as predefined algebraic structures. We move to constructing one from scratch. That is, we define a vector space of $\mathbb{C}$-linear combinations of basis elements e_i i.e

$$V = \left\{ \sum \lambda_i \cdot e_i \mid \lambda_i \in \mathbb{C} \right\}$$

Then, we may define an $n-$dimensional representation by

$$\rho : G \to GL_n(V)$$

($GL_n(V)$) represents group of invertible linear transformations from $V \to V$) via

$$\rho(g)(\sum_{i=1}^{n} \lambda_i e_i) = \sum_{i=1}^{n} \lambda_i (g \cdot e_i)$$

In short, we're essentially translating our basis by action of g.

Remark 2. $GL_n(V) \cong GL_N(\mathbb{C})$

Example 3. Let S_n be the symmetric group. Let $S = \{e_i, \ldots, e_n\}$ be a finite set of

elements. Then

$$g \cdot e_i = e_{g(i)}$$

is a permutation representation of S_n.

Example 4. Consider the symmetric group S_3. We have that

$$\rho((12)) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\rho((123)) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\rho((23)) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Definition 3. Two representations

$$\rho_1 : G \to GL_m(\mathbb{C})$$
$$\rho_2 : G \to GL_n(\mathbb{C})$$

are ***isomorphic*** if $m = n$ and there exists $A \in GL_n(\mathbb{C})$ such that

$$\rho_2(x) = A\rho_1(x)A^{-1}$$

for all $x \in G$.

Example 5. Consider

Definition 4. A representation $\rho : G \to GL_n(V)$ is ***irreducible***, if there does not exist a non-zero subspace $W \subset V$ such that

$$\rho(x)W \subseteq W$$

for all $x \in G$. W is known as a ***subrepresentation***.